

Hemostasis and Thrombosis

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2.1

Coagulation and Fibrinolytic Systems/Reagents and Methods

1. Which of the following initiates in vivo coagulation by activation of factor VII?

- A. Protein C
- B. Tissue factor (TF)
- C. Plasmin activator
- D. Thrombomodulin (TM)

Hemostasis/Apply knowledge of fundamental biological characteristics/Coagulation/2

Answer to Question 1

- 1. B** In vivo, activation of coagulation occurs on the surface of activated platelets (PLTs) or cells that have TF. TF is found on the surface of many cells outside the vascular system (extrinsic). On vascular injury, TF is exposed to the vascular system. TF has high affinity for factors VII and VIIa. TF activates factor VII to VIIa and forms TF–VIIa complex. TF–VIIa complex in the presence of calcium ion (Ca^{2+}) and PLT phospholipid activates factor IX to factor IXa and factor X to factor Xa. Factor Xa forms a complex with cofactor Va (Xa–Va) on the surface of the activated PLTs. Factor Xa–Va complex in the presence of Ca^{2+} and PLT phospholipid converts prothrombin (factor II) to thrombin (IIa). Thrombin acts on soluble plasma fibrinogen to form a fibrin clot, which is stabilized by activated factor XIII (XIIIa). In addition, activated factor IX (IXa) forms a complex with activated cofactor VIII (VIIIa) on the surface of the activated PLTs. Factor IXa–VIIIa complex in the presence of Ca^{2+} and PLT phospholipid converts factor X to Xa with the end products of thrombin and fibrin clot, as discussed previously. The classical description of intrinsic, extrinsic, and common pathways does not take place in vivo. The concept of these three pathways is used to explain clot formation in laboratory tests. Activated thromboplastin time (APTT) is determined by the intrinsic and common pathways, whereas prothrombin time (PT) is determined by the extrinsic and common pathways. The extrinsic pathway is so named because TF is derived from extravascular cells.

2. Which of the following clotting factors plays a role in clot formation in vitro, but not in vivo?

- A. Factor VIIa
- B. Factor IIa
- C. Factor XIIIa
- D. Factor Xa

Hemostasis/Apply knowledge of fundamental biological characteristics/Coagulation/2

3. The anticoagulant of choice for most routine coagulation studies is:

- A. Sodium oxalate
- B. Sodium citrate
- C. Heparin
- D. Ethylenediaminetetraacetic acid (EDTA)

Hemostasis/Select methods/Reagents/Specimen collection and handling/Specimen/1

4. Which anticoagulant:blood ratio is correct for coagulation procedures?

- A. 1:4
- B. 1:5
- C. 1:9
- D. 1:10

Hemostasis/Select methods/Reagents/Specimen collection and handling/Specimen/1

5. Which results would be expected for PT and APTT in a patient with polycythemia?

- A. Both prolonged
- B. Both shortened
- C. Normal PT, prolonged APTT
- D. Both normal

Hemostasis/Correlate clinical and laboratory data/Coagulation tests/3

6. Which reagents are used in the PT test?

- A. Thromboplastin and sodium chloride
- B. Thromboplastin and potassium chloride
- C. Thromboplastin and calcium
- D. Actin and calcium chloride

Hemostasis/Select methods/Reagents/Coagulation tests/1

7. Which test would be abnormal in a patient with factor X deficiency?

- A. PT only
- B. APTT only
- C. PT and APTT
- D. Thrombin time (TT)

Hemostasis/Correlate clinical and laboratory data/Coagulation tests/2

Answers to Questions 2–7

- 2. **C** Factor XIIa does not play a role in coagulation in vivo; however, in vitro, the deficiency of this factor causes prolonged APTT. In vitro, factor XII is activated by substances, such as glass, Kaolin, and ellagic acid, and in vivo it may be activated by exposure to a negatively charged cell surface membrane, such collagen, as well as to kallikrein and high-molecular-weight kininogen (HMWK). In vivo, factor XIIa plays an important role in the fibrinolytic system by activating plasminogen to plasmin. Plasmin degrades the fibrin clot at the site of injury. Deficiency of factor XII is associated with thrombosis and not with bleeding. Factors VIIa, Xa, and IIa play a role in vivo and in vitro.
- 3. **B** The anticoagulant of choice for most coagulation procedures is sodium citrate (3.2%). Because factors V and VIII are more labile in sodium oxalate, heparin neutralizes thrombin, and EDTA inhibits thrombin's action on fibrinogen, these anticoagulants are not used for routine coagulation studies.

4. **C** The optimal anticoagulant: blood ratio is one part anticoagulant to nine parts blood. The anticoagulant supplied in this amount is sufficient to bind all the available calcium, thereby preventing clotting.
5. **A** The volume of blood in a polycythemic patient contains so little plasma that excess anticoagulant remains and is available to bind to reagent calcium, thereby resulting in prolongation of PT and APTT. For more accurate results, the plasma: anticoagulant ratio can be modified by decreasing the amount of anticoagulant in the collection tube by using the following formula: $(0.00185)(V)(100-H) = C$, where V = blood volume in milliliters (mL); H = patient's hematocrit (Hct); and C = volume (mL) of anticoagulant. A new sample should be drawn to rerun the PT and APTT tests.
6. **C** Thromboplastin and calcium (combined into a single reagent) replace tissue thromboplastin and calcium necessary, in vivo, to activate factor VII to factor VIIa. This ultimately generates thrombin from prothrombin via the coagulation cascade.
7. **C** Factor X is involved in the common pathway of the coagulation cascade; therefore, its deficiency prolongs both PT and APTT. Activated factor X along with activated factor V in the presence of calcium and platelet factor 3 (PF3) converts prothrombin (factor II) to the active enzyme thrombin (factor IIa).

8. Which clotting factor is *not* measured by the PT and APTT tests?

- A. Factor VIII
- B. Factor IX
- C. Factor V
- D. Factor XIII

Hemostasis/Apply principles of basic laboratory procedures/Coagulation tests/1

9. A modification of which procedure can be used to measure fibrinogen?

- A. PT
- B. APTT
- C. TT
- D. Fibrin degradation products

Hemostasis/Apply principles of basic laboratory procedures/Coagulation tests/2

10. Which of the following characterizes vitamin K?

- A. It is required for biological activity of fibrinolysis
- B. Its activity is enhanced by heparin therapy
- C. It is required for carboxylation of glutamate residues of some coagulation factors
- D. It is made by endothelial cells

Hemostasis/Apply knowledge of fundamental biological characteristics/Vitamin K/2

11. Which fragments of fibrin clot degradation are measured by the D-dimer test?

- A. Fragments X and Y
- B. Fibrinopeptide A and B
- C. Fragments D and E
- D. The D-D domains

Hemostasis/Apply principles of basic laboratory procedures/D-dimer/2

12. Which of the following clotting factors are measured by the APTT test?

- A. Factors II, VII, IX, X

- B. Factors VII, X, V, II, I
- C. Factors XII, XI, IX, VIII, X, V, II, I
- D. Factors XII, VII, X, V, II, I

Hemostasis/Correlate clinical and laboratory data/Coagulation tests/2

13. Which coagulation test(s) would be abnormal in a patient with vitamin K deficiency?

- A. PT only
- B. PT and APTT
- C. Fibrinogen level
- D. TT

Hemostasis/Correlate clinical and laboratory data/Coagulation tests/2

Answers to Questions 8–14

- 8. **D** Factor XIII is not measured by PT or APTT. Factor XIII (fibrin stabilizing factor) is a transamidase. It creates covalent bonds between fibrin monomers, which are formed during the coagulation process, to produce a stable fibrin clot. In the absence of factor XIII, the hydrogen bonded fibrin polymers are unstable and, therefore, soluble in 5M urea or in 1% monochloroacetic acid.
- 9. **C** Fibrinogen can be quantitatively measured by modification of the TT by diluting the plasma because the thrombin clotting time of diluted plasma is inversely proportional to the concentration of fibrinogen (principle of Clauss method).
- 10. **C** Vitamin K is necessary for activation of vitamin K–dependent clotting factors (II, VII, IX, and X). This activation is accomplished by carboxylation of glutamic acid residues of the inactive clotting factors. The activity of vitamin K is not enhanced by heparin therapy. Vitamin K is present in a variety of foods and is also the only vitamin made by the organisms living in the intestine.
- 11. **D** D-dimer is a specific product resulting from digestion of cross-linked fibrin only. It consists of two D domains called D-D fragment and is a marker for thrombosis and fibrinolysis. In the D-dimer test, monoclonal antibody is directed against D-dimer antigen by using a variety of automated instruments. The D-dimer test is used to diagnose acute and chronic disseminated intravascular coagulation (DIC) and to rule out thromboembolic disorders. Fragments X, Y, D, and E are produced by the action of plasmin on fibrinogen and fibrin. Polypeptide A and B are produced by the proteolytic action of thrombin on fibrinogen to form a fibrin clot.
- 12. **C** The APTT test evaluates the clotting factors in the intrinsic pathway (XII, XI, IX, and VIII) as well as those in the common pathway (X, V, II, and I).
- 13. **B** Patients with vitamin K deficiency exhibit decreased production of functional prothrombin proteins (factors II, VII, IX, and X). Decreased levels of these factors prolong both PT and APTT.

14. Which of the following is correct regarding the international normalized ratio (INR)?

- A. It uses the international sensitivity ratio (ISR)
- B. It standardizes PT results
- C. It standardizes APTT results

D. It is used to monitor heparin therapy

Hemostasis/Apply knowledge of fundamental biological characteristics/INR/2

15. Which of the following is referred to as an endogenous activator of plasminogen?

A. Streptokinase

B. Transamidase

C. Tissue plasminogen activator (tPA)

D. tPA inhibitor

Hemostasis/Apply knowledge of fundamental biological characteristics/Plasminogen/2

16. Which protein is the primary inhibitor of the fibrinolytic system?

A. Protein C

B. Protein S

C. α_2 -Antiplasmin

D. α_2 -Macroglobulin

Hemostasis/Apply knowledge of fundamental biological characteristics/Plasmin/1

17. Which of the following statements is correct regarding the D-dimer test?

A. Levels are decreased in DIC

B. Test detects polypeptides A and B

C. Test detects fragments D and E

D. Test has a negative predictive value

Hemostasis/Apply principles of basic laboratory procedures/D-dimer/2

18. A protein that plays a role in both coagulation and PLT aggregation is:

A. Factor I

B. Factor VIII

C. Factor IX

D. Factor XI

Hemostasis/Apply knowledge of fundamental biological characteristics/Clotting factors/2

19. A standard 4.5-mL blue-top tube filled with 3.0 mL of blood was submitted to the laboratory for PT and APTT tests. The sample is from a patient undergoing surgery the following morning for a tonsillectomy. Which of the following is the necessary course of action by the medical laboratory scientist?

A. Run both tests in duplicate and report the average result

B. Reject the sample and request a new sample

C. Report the PT result

D. Report the APTT result

Hemostasis/Select methods/Reagents/Specimen collection and handling/Specimens/3

Answers to Questions 14–19

14. B INR is used to standardize PT results to adjust for the differences in thromboplastin reagents made by different manufacturers and used by various institutions. The INR calculation uses the international sensitivity index (ISI) value and is used to monitor an oral anticoagulant, such as warfarin. INR is not used to standardize APTT testing.

15. C tPA is an endogenous (produced in the body) activator of plasminogen. It is released from the endothelial cells by the action of protein C. It converts plasminogen to

plasmin. Streptokinase is an exogenous (not made in the body) activator of plasminogen.

16. **C** α_2 -Antiplasmin is the main inhibitor of plasmin. It inhibits plasmin by forming a 1:1 stoichiometric complex with any free plasmin in plasma and, thus, prevents binding of plasmin to fibrin and fibrinogen.
17. **D** The D-dimer assay evaluates fibrin degradation. It is a nonspecific screening test that shows increased values in many conditions in which fibrinolysis is increased, such as DIC and fibrinolytic therapy. The D-dimer test is widely used to rule out thrombosis and thrombotic activities. The negative predictive value of a test is the probability that a person with a negative result is free of the disease that the test is meant to detect. Therefore, a negative D-dimer test result rules out thrombosis, and further laboratory investigations are not required.
18. **A** Factor I (fibrinogen), along with the glycoprotein IIb–IIIa complex, is necessary for PLT aggregation. Factor I is also a substrate in the common pathway of coagulation. Thrombin acts on fibrinogen to form fibrin clots.
19. **B** A 4.5-mL blue-top tube contains 4.5 mL blood + 0.5 mL sodium citrate. The tube should be 90% full. A tube with 3.0 mL blood should be rejected because the quantity is not sufficient (QNS). QNS samples alter the necessary blood:anticoagulant ratio (9:1). The excess anticoagulant in a QNS sample binds to the reagent calcium, resulting in prolongation of PT and APTT.

20. Which statement is correct regarding sample storage for the PT test?

- A. Stable for 24 hours if the sample is capped
- B. Stable for 24 hours if the sample is refrigerated at 4°C
- C. Stable for 4 hours if the sample is stored at 4°C
- D. Should be run within 8 hours

Hemostasis/Select methods/Reagents/Specimen collection and handling/Specimens/2

21. In primary fibrinolysis, the fibrinolytic activity results in response to:

- A. Increased fibrin formation
- B. Spontaneous activation of fibrinolysis
- C. Increased fibrin monomers
- D. DIC

Hemostasis/Apply knowledge of fundamental biological characteristics/Fibrinolysis/2

22. Plasminogen deficiency is associated with:

- A. Bleeding
- B. Thrombosis
- C. Increased fibrinolysis
- D. Increased coagulation

Hemostasis/Correlate clinical and laboratory data/Plasminogen/2

23. Which of the following clotting factors are activated by thrombin that is generated by tissue pathway (TF-VIIa)?

- A. Factors XII, XI
- B. Factors XII, I
- C. Factors I, II

D. Factors V, VIII

Hemostasis/Apply knowledge of fundamental biological characteristics/Thrombin/2

24. Which substrate is used in a chromogenic factor assay?

- A. p-nitroaniline (pNa)**
- B. Chlorophenol red**
- C. Prussian blue**
- D. Ferricyanide**

Hemostasis/Selected methods/Reagents/Chromogenic assays/1

25. Which of the following antibodies is used in the D-dimer assay?

- A. Polyclonal antibody directed against X and Y fragments**
- B. Polyclonal antibody directed against D-dimer**
- C. Monoclonal antibody against D and E fragments**
- D. Monoclonal antibody against D-dimer**

Hemostasis/Selected methods/Reagents/D-dimer assay/2

Answers to Questions 20–25

- 20. A** According to Clinical Laboratory Standards Institute (CLSI) guidelines, plasma samples for PT testing, if capped, are stable for 24 hours at room temperature. Refrigerating the sample causes cold activation of factor VII and, therefore, shortened PT results. APTT samples are stable for 4 hours if stored at 4°C.
- 21. B** Primary fibrinolysis is a rare pathological condition in which spontaneous systemic fibrinolysis occurs. Plasmin is formed in the absence of coagulation activation and clot formation. Primary fibrinolysis is associated with increased production of plasminogen and plasmin, decreased plasmin removal from the circulation, and spontaneous bleeding.
- 22. B** Plasminogen deficiency is associated with thrombosis. Plasminogen is an important component of the fibrinolytic system. Plasminogen is activated to plasmin, which is necessary for the degradation of fibrin clots to prevent thrombosis. When plasminogen is deficient, plasmin is not formed, causing a defect in the clot lysing processes.
- 23. D** Factors V and VIII are activated by the thrombin that is generated by the action of TF-VIIIa on factor X to form factor Xa. Factor Xa forms a complex with factor Va on the PLT surfaces. Factor Xa–Va complex in the presence of phospholipid and Ca^{2+} transform more prothrombin to thrombin.
- 24. A** The chromogenic, or amidolytic, assays use a color-producing substance, known as *chromophore*. The chromophore used for the coagulation laboratory is pNa. pNa is bound to a synthetic oligopeptide substrate. The protease cleaves the chromogenic substrate at the site binding the oligopeptide to the pNa, which results in release of pNa. Free pNa has a yellow color; the color intensity of the solution is proportional to protease activity and is measured by a photodetector at 405 nm.
- 25. D** The D-dimer is the fibrin degradation product generated by the action of plasmin on cross-linked fibrin formed by XIIIa. The patient's plasma is mixed with latex particles coated with monoclonal antibodies against D-domains. The test can be automated, or performed manually on a glass slide by looking macroscopically for agglutination.

Enzyme-linked immunosorbent assay (ELISA) methods are also available. Normal D-dimer in plasma is less than 2 ng/mL. Increased levels of D-dimer are associated with DIC, thrombolytic therapy, venous thrombosis, and thromboembolic disorders. The D-dimer assay has a 90% to 95% negative predictive value and has been used to rule out thrombosis and thromboembolic disorders.

2.2

Platelet and Vascular Disorders

1. Thrombotic thrombocytopenic purpura (TTP) is characterized by:

- A. Prolonged PT
- B. Increased PLT aggregation
- C. Thrombocytosis
- D. Prolonged APTT

Hemostasis/Correlate clinical and laboratory data/Platelets/2

2. Thrombocytopenia may be associated with:

- A. Splenectomy
- B. Hypersplenism
- C. Acute blood loss
- D. Increased proliferation of pluripotent stem cells

Hemostasis/Apply knowledge of fundamental biological characteristics/Platelets/2

3. Aspirin prevents PLT aggregation by inhibiting the action of which enzyme?

- A. Phospholipase
- B. Cyclo-oxygenase
- C. Thromboxane A₂ (TXA₂) synthetase
- D. Prostacyclin synthetase

Hemostasis/Apply knowledge of fundamental biological characteristics/Platelets/1

4. Normal PLT adhesion depends on:

- A. Fibrinogen
- B. Glycoprotein Ib
- C. Glycoprotein IIb–IIIa complex
- D. Calcium

Hemostasis/Apply knowledge of fundamental biological characteristics/Platelets/1

5. Which of the following test results is normal in a patient with classic von Willebrand disease?

- A. PLT aggregation
- B. APTT
- C. PLT count
- D. Factor VIII:C and von Willebrand factor (VWF) levels

Hemostasis/Correlate clinical and laboratory data/Platelet disorders/2

6. Bernard–Soulier syndrome is associated with:

- A. Decreased factor IX
- B. Decreased factor VIII
- C. Thrombocytopenia and giant PLTs
- D. Abnormal PLT function test results

Hemostasis/Correlate clinical and laboratory data/Platelet disorders/2

Answers to Questions 1–6

1. **B** TTP is a quantitative PLT disorder associated with increased intravascular PLT activation and aggregation resulting in thrombocytopenia. PT and APTT results are normal in TTP.
 2. **B** Hypersplenism is associated with thrombocytopenia. In this condition, up to 90% of PLTs can be sequestered in the spleen, causing decreases in circulatory PLTs. Splenectomy, acute blood loss, and increased proliferation of pluripotent stem cells are associated with thrombocytosis.
 3. **B** Aspirin prevents PLT aggregation by inhibiting the activity of the enzyme cyclo-oxygenase. This inhibition prevents the formation of TXA₂, which serves as a potent PLT aggregator.
 4. **B** Glycoprotein Ib is a PLT receptor for VWF. Glycoprotein Ib and VWF are both necessary for a normal PLT adhesion. Other proteins that play a role in PLT adhesion are glycoproteins V and IX.
 5. **C** Von Willebrand disease is an inherited, qualitative PLT disorder that results in increased bleeding, prolonged APTT, and decreased factor VIII:C and VWF levels. The PLT count and morphology are generally normal in von Willebrand disease, but PLT aggregation in the PLT function assay is abnormal.
 6. **C** Bernard–Soulier syndrome is associated with thrombocytopenia and giant PLTs. It is a qualitative PLT disorder caused by the deficiency of glycoprotein Ib. In Bernard–Soulier syndrome, PLT aggregation in the PLT function assay is abnormal. Factor VIII and IX assays are not indicated for this diagnosis.
7. **When performing PLT aggregation studies, which set of PLT aggregation results would most likely be associated with Bernard–Soulier syndrome?**
- A. Normal PLT aggregation to collagen, adenosine diphosphate (ADP), and ristocetin
 - B. Normal PLT aggregation to collagen, ADP, and epinephrine (EPI); decreased aggregation to ristocetin
 - C. Normal PLT aggregation to EPI and ristocetin; decreased aggregation to collagen and ADP
 - D. Normal PLT aggregation to EPI, ristocetin, and collagen; decreased aggregation to ADP
- Hemostasis/Correlate clinical and laboratory data/Platelet disorders/2*
8. **Which set of PLT responses would be most likely associated with Glanzmann thrombasthenia?**
- A. Normal PLT aggregation to ADP and ristocetin; decreased aggregation to collagen
 - B. Normal PLT aggregation to collagen; decreased aggregation to ADP and ristocetin
 - C. Normal PLT aggregation to ristocetin; decreased aggregation to collagen, ADP, and EPI
 - D. Normal PLT aggregation to ADP; decreased aggregation to collagen and ristocetin
- Hemostasis/Correlate clinical and laboratory data/Platelet disorders/2*
9. **Which of the following is a characteristic of acute immune thrombocytopenic purpura?**
- A. Spontaneous remission within a few weeks
 - B. Predominantly seen in adults
 - C. Nonimmune PLT destruction

D. Insidious onset

Hemostasis/Apply knowledge of fundamental biological characteristics/Platelet disorders/2

10. TTP differs from DIC in that:

- A.** APTT is normal in TTP but prolonged in DIC
- B.** Schistocytes are not present in TTP but are present in DIC
- C.** PLT count is decreased in TTP but normal in DIC
- D.** PT is prolonged in TTP but decreased in DIC

Hemostasis/Correlate clinical and laboratory data/Platelet disorders/2

11. Several hours after birth, a baby boy develops petechiae and purpura and hemorrhagic diathesis. The PLT count is $18 \times 10^9/L$. What is the most likely explanation for the low PLT count?

- A.** Drug-induced thrombocytopenia
- B.** Secondary thrombocytopenia
- C.** Neonatal alloimmune thrombocytopenia
- D.** Neonatal DIC

Hemostasis/Correlate clinical and laboratory data/Platelet disorders/3

Answers to Questions 7–11

- 7. B** Bernard–Soulier syndrome is a disorder of PLT adhesion caused by deficiency of glycoprotein Ib. PLT aggregation is normal in response to collagen, ADP, and EPI but abnormal in response to ristocetin.
- 8. C** Glanzmann thrombasthenia is a disorder of PLT aggregation. PLT aggregation is normal in response to ristocetin, but abnormal in response to collagen, ADP, and EPI.
- 9. A** Acute immune thrombocytopenic purpura is an immune-mediated disorder found predominantly in children. It is commonly associated with infection (primarily viral). It is characterized by abrupt onset, and spontaneous remission usually occurs within several weeks.
- 10. A** TTP is a PLT disorder in which PLT aggregation increases, resulting in thrombocytopenia. Schistocytes are present in TTP as a result of microangiopathic hemolytic anemia (MAHA); however, the PT and APTT are both normal. In DIC, the PT and APTT are both prolonged, the PLT count is decreased, and schistocytes are seen in the peripheral blood smear.
- 11. C** Neonatal alloimmune thrombocytopenia is similar to the hemolytic disease of the fetus and newborn. It results from immunization of the mother by fetal PLT antigens. The offending antibodies are commonly anti-PLT antigen A¹ (PlA¹), also referred to as human platelet antigen (HPA) 1a. These alloantibodies are directed against glycoproteins IIb/IIIa, Ib/IX, Ia/IIb, and CD 109. Maternal antibodies cross the placenta, resulting in thrombocytopenia in the fetus.

12. Which of the following is associated with post-transfusion purpura (PTP)?

- A.** Nonimmune thrombocytopenia/alloantibodies
- B.** Immune-mediated thrombocytopenia/alloantibodies
- C.** Immune-mediated thrombocytopenia/autoantibodies

D. Nonimmune-mediated thrombocytopenia/autoantibodies

Hemostasis/Apply knowledge of fundamental biological characteristics/Platelet disorders/2

13. Hemolytic uremic syndrome (HUS) is associated with:

A. Fever, thrombocytosis, anemia, and renal failure

B. Fever, granulocytosis, and thrombocytosis

C. *Escherichia coli* 0157:H7

D. Leukocytosis and thrombocytosis

Hemostasis/Apply knowledge of fundamental biological characteristics/Platelet disorders/2

14. Storage pool deficiencies are defects of:

A. PLT adhesion

B. PLT aggregation

C. PLT granules

D. PLT production

Hemostasis/Apply knowledge of fundamental biological characteristics/Platelet disorders/1

15. Lumi-aggregation measures:

A. PLT aggregation only

B. PLT aggregation and adenosine triphosphate (ATP) release

C. PLT adhesion

D. PLT glycoprotein Ib

Hemostasis/Select methods/Reagents/Specimen collection and handling/Aggregometry/1

16. Neurological findings may be commonly associated with which of the following disorders?

A. HUS

B. TTP

C. ITP

D. PTP

Hemostasis/Apply knowledge of fundamental biological characteristics/Platelet function/1

17. Which of the following is correct regarding acquired TTP?

A. Autoimmune disease

B. Decreased VWF

C. Decreased PLT aggregation

D. Decreased PLT adhesion

Hemostasis/Apply knowledge of fundamental biological characteristics/Platelet disorders/2

18. Hereditary hemorrhagic telangiectasia is a disorder of:

A. PLTs

B. Clotting proteins

C. Fibrinolysis

D. Connective tissue

Hemostasis/Apply knowledge of fundamental biological characteristics/2

Answers to Questions 12–18

- 12. B** PTP is a rare form of alloimmune thrombocytopenia characterized by severe thrombocytopenia occurring after transfusion of blood or blood products. PTP is caused by antibody-related PLT destruction in previously immunized patients. In the

majority of cases, the alloantibody produced is against Pl^{A1} (HPA-1a).

- 13. **C** HUS is caused by *E. coli* 0157:H7. It is associated with ingestion of *E. coli*-contaminated foods and is commonly seen in children. The clinical manifestations in HUS are fever, diarrhea, thrombocytopenia, MAHA, and renal failure.
- 14. **C** Storage pool deficiencies are defects of PLT granules. Most commonly, a decrease in PLT-dense granules is present with decreased release of ADP, ATP, calcium, and serotonin from PLT-dense granules.
- 15. **B** Lumi-aggregation measures PLT aggregation and ATP release. It is performed on whole blood diluted with saline. PLT aggregation is measured by impedance, whereas ATP release is measured by addition of luciferin to a blood sample. There is no ATP release in storage pool deficiencies.
- 16. **B** TTP is characterized by neurological problems, fever, thrombocytopenia, MAHA, and renal failure.
- 17. **A** Acquired TTP is an autoimmune disease associated with autoantibodies produced against VWF cleaving enzyme (ADAMTS-13). This deficiency results in an increase in plasma VWF and consequently increased PLT aggregation and thrombocytopenia.
- 18. **D** Hereditary hemorrhagic telangiectasia (Osler–Weber–Rendu syndrome) is a connective tissue disorder associated with telangiectases (dilated capillaries) of the mucous membranes and skin. Lesions may develop on the tongue, lips, palate, face, hands, and nasal mucosa and throughout the gastrointestinal tract. This disorder is an autosomal dominant condition that usually manifests in adolescence or early adulthood.

19. Which of the following prevents PLT aggregation?

- A. TXA2
- B. Thromboxane B_2
- C. Prostacyclin
- D. Antithrombin (AT)

Hemostasis/Apply knowledge of fundamental biological characteristics/Platelets/2

20. Which defect characterizes Gray syndrome?

- A. PLT adhesion defect
- B. Dense granule defect
- C. Alpha granule defect
- D. Coagulation defect

Hemostasis/Apply knowledge of fundamental biological characteristics/Platelet disorders/2

21. The P2Y₁₂ ADP receptor agonist assay may be used to monitor PLT aggregation inhibition to which of the following drugs?

- A. Warfarin
- B. Heparin
- C. Low-molecular-weight heparin (LMWH)
- D. Clopidogrel (Plavix)

Hemostasis/Selected methods/Reagents/Special tests/2

Answers to Questions 19–21

- 19. C** Prostacyclin is released from the endothelium and is an inhibitor of PLT aggregation. TXA₂ promotes PLT aggregation. Thromboxane B₂ is an oxidized form of TXA₂ and is excreted in urine. AT is a physiological anticoagulant.
- 20. C** Gray syndrome is a PLT granule defect associated with a decrease in alpha granules resulting in decreased production of alpha granule proteins, such as PF4 and beta thromboglobulin. Alpha granule deficiency results in the appearance of agranular PLTs when viewed in a Wright-stained blood smear.
- 21. D** The VerifyNow P2Y₁₂ test is used to assess a patient's response to antiplatelet drugs, such as clopidogrel (Plavix) and prasugrel (Effient). These drugs are given orally along with aspirin for prevention of thrombosis or as alternative antiplatelet drugs for patients who cannot tolerate or are not sensitive to aspirin. Clopidogrel and prasugrel prevent PLT aggregation by irreversibly binding to P2Y₁₂, which is a PLT membrane receptor for ADP. The VerifyNow P2Y₁₂ test is a whole blood test and uses ADP as an aggregating agent to measure the level of PLT aggregation impaired by these medications. The baseline value for PLT aggregation is established. The percent (%) change from baseline aggregation is calculated and reported as % P2Y₁₂ inhibition.

22. Which of the following instruments can be used to evaluate PLT function?

- A. PLT aggregometer
- B. VerifyNow
- C. PFA-100
- D. All of the above

Hemostasis/Selected methods/Reagents/Special tests/2

23. Which of the following PLT aggregating agents demonstrates a monophasic aggregation curve when used in the optimal concentration?

- A. Thrombin
- B. Collagen
- C. ADP
- D. EPI

Hemostasis/Apply knowledge of fundamental biological characteristics/Aggregating agents/1

Answers to Questions 22–23

- 22. D** All of the instruments listed can be used to evaluate PLT function. PLT function testing is done either to determine the cause of bleeding in a patient with normal PLT count and normal coagulation tests or to assess the efficacy of antiplatelet drugs. PLT aggregometry is used for the diagnosis of inherited PLT disorders. A PLT aggregometer uses PLT-rich plasma to measure PLT aggregation in response to different PLT aggregating agents by measuring light transmission. A Lumi-aggregometer uses whole blood and has the ability to measure dense-granule secretion (by using a luminescent marker) in addition to PLT aggregation. The VerifyNow measures a patient's response to multiple antiplatelet drugs, including aspirin, P2Y₁₂ inhibitors, and glycoprotein IIb/IIIa inhibitors. The Platelet Function Analyzer-100

(PFA-100) is used as a screening tool for PLT function, and has replaced the bleeding time test. It uses citrated whole blood and two cartridges. Whole blood (800 μ L) is put into each test cartridge. Vacuum is used to move the blood through a very thin glass tube that has been coated with a membrane containing collagen and either EPI or ADP. This coating activates the PLTs in the moving sample and promotes PLT adhesion and aggregation. The time it takes for the clot to form inside the glass tube and prevent further flow is measured as the closure time (CT). Initial screening is done with collagen/EPI. If CT is normal, it is unlikely that PLT dysfunction exists. The collagen/ADP membrane is used to confirm an abnormal collagen/EPI test result. If both tests show abnormal results, it is likely that the patient has a PLT dysfunction, and further testing for inherited and acquired bleeding disorders is indicated. If collagen/ADP is normal, then the abnormal collagen/EPI test result is likely caused by aspirin sensitivity.

- 23. B** Collagen is the only commonly used agent that demonstrates a single-wave (monophasic) response preceded by a lag time.

2.3

Coagulation System Disorders

1. The APTT is sensitive to a deficiency of which clotting factor?

- A. Factor VII
- B. Factor X
- C. PF3
- D. Calcium

Hemostasis/Evaluate laboratory data to recognize health and disease states/Factor deficiency/2

2. Which test result would be normal in a patient with dysfibrinogenemia?

- A. TT
- B. APTT
- C. PT
- D. Immunologic fibrinogen level

Hemostasis/Correlate clinical and laboratory data/Factor deficiency/2

3. A patient with a prolonged PT is given intravenous vitamin K. PT is corrected to normal after 24 hours. Which clinical condition most likely caused these results?

- A. Necrotic liver disease
- B. Factor X deficiency
- C. Fibrinogen deficiency
- D. Obstructive jaundice

Hemostasis/Correlate clinical and laboratory data/Vitamin K deficiency/3

4. Which factor deficiency is associated with prolonged PT and APTT?

- A. Factor X
- B. Factor VIII
- C. Factor IX
- D. Factor XI

Hemostasis/Evaluate laboratory data to recognize health and disease states/Factor deficiency/2

5. Prolonged APTT is corrected with factor VIII–deficient plasma but not with factor IX–deficient plasma. Which factor is deficient?

- A. Factor V
- B. Factor VIII
- C. Factor IX
- D. Factor X

Hemostasis/Evaluate laboratory data to recognize health and disease states/Factor deficiency/3

6. Which of the following is a characteristic of classic hemophilia A?

- A. Abnormal PLT aggregation
- B. Autosomal recessive inheritance

- C. Mild to severe bleeding episodes
- D. Prolonged PT

Hemostasis/Correlate clinical and laboratory data/Hemostasis/Hemophilia/2

Answers to Questions 1–6

1. **B** The APTT is sensitive to the deficiency of coagulation factors in the intrinsic pathway (factors XII, XI, IX, and VIII) and the common pathway (factors X, V, II, and I).
2. **D** The level of plasma fibrinogen determined immunologically is normal. In a patient with dysfibrinogenemia, fibrinogen is not polymerized properly, causing abnormal results of the fibrinogen-dependent coagulation tests.
3. **D** Obstructive jaundice contributes to coagulation disorders by preventing vitamin K absorption. Vitamin K is fat soluble and requires bile salts for absorption. Parenteral administration of vitamin K bypasses the bowel, hence the need for bile salts.
4. **A** Factor X, a common pathway factor deficiency, is most likely suspected because both PT and APTT are prolonged. Other causes may include liver disease, vitamin K deficiency, and anticoagulant drugs, such as warfarin and heparin.
5. **C** Because prolonged APTT is not corrected with factor IX–deficient plasma, factor IX is suspected to be deficient in the test plasma.
6. **C** Hemophilia A (factor VIII deficiency) is characterized by mild to severe bleeding episodes, depending on the concentration of factor VIII:C. Hemophilia A is inherited as a sex-linked disease. PLT aggregation and PT are both normal in hemophilia A.

7. Refer to the following results:

PT = prolonged

APTT = prolonged

PLT count = decreased

Which disorder may be indicated?

- A. Factor VIII deficiency
- B. von Willebrand disease
- C. DIC
- D. Factor IX deficiency

Hemostasis/Correlate clinical and laboratory data/Coagulation disorders/3

8. Which of the following is a predisposing condition for the development of DIC?

- A. Adenocarcinoma
- B. Idiopathic thrombocytopenic purpura (ITP)
- C. Post transfusion purpura (PTP)
- D. Heparin-induced thrombocytopenia (HIT)

Hemostasis/Correlate clinical and laboratory data/DIC/1

9. Factor XII deficiency is associated with:

- A. Bleeding episodes
- B. Epistaxis
- C. Decreased risk of thrombosis
- D. Increased risk of thrombosis

Hemostasis/Apply knowledge of fundamental biological characteristics/Factor deficiency/2

10. The following results were obtained on a patient: normal PLT count and function, normal PT, and prolonged APTT. Which of the following disorders is most consistent with these results?

- A. Hemophilia A
- B. Bernard–Soulier syndrome
- C. von Willebrand disease
- D. Glanzmann thrombasthenia

Hemostasis/Correlate clinical and laboratory data/Coagulation disorders/3

11. The following laboratory results were obtained from a 40-year-old woman: PT = 20 sec; APTT = 50 sec; TT = 18 sec. What is the most probable diagnosis?

- A. Factor VII deficiency
- B. Factor VIII deficiency
- C. Factor X deficiency
- D. Hypofibrinogenemia

Hemostasis/Correlate clinical and laboratory data/Factor deficiency/3

12. When performing a factor VIII activity assay, a patient's plasma is mixed with:

- A. Normal patient plasma
- B. Factor VIII–deficient plasma
- C. Plasma with a high concentration of factor VIII
- D. Normal control plasma

Hemostasis/Apply principles of basic laboratory procedures/Factor assay/2

Answers to Questions 7–12

- 7. **C** In DIC, there is a diffuse intravascular generation of thrombin and fibrin. As a result, clotting factors and PLTs are consumed, resulting in decreased PLT count and increased PT and APTT.
- 8. **A** Adenocarcinoma can liberate procoagulant (thromboplastic) substances that can activate prothrombin intravascularly. ITP is a thrombocytopenia caused by an autoantibody; PTP is an alloimmune thrombocytopenia caused by transfusion of blood or blood products; HIT results from an antibody to heparin–PF4 complex causing thrombocytopenia in 1% to 5% of patients who are on heparin therapy. In some affected persons, thrombosis may also occur.
- 9. **D** Factor XII–deficient patients commonly have thrombotic episodes. In vitro, activated factor XII (XIIa) activates factor XI to factor XIa. In vivo, however, factor XII plays a major role in the fibrinolytic system by activating plasminogen to form plasmin. Hemorrhagic manifestations are not associated with factor XII deficiency because VIIa/TF can activate factor IX to factor IXa and factor X to factor Xa to generate thrombin.
- 10. **A** Hemophilia A is associated with deficiency of factor VIII, resulting in bleeding and abnormal APTT. The PLT number and function are normal in this disorder. Von Willebrand disease is a disorder of PLT adhesion associated with decreased VWF and factor VIII, causing an abnormal PLT function test result and abnormal APTT. Both

Glanzmann thrombasthenia and Bernard–Soulier syndrome cause abnormal PLT aggregation but do not cause abnormal APTT.

- 11. D** Fibrinogen (factor I) is a clotting protein of the common pathway and is evaluated by TT. In hypofibrinogenemia (fibrinogen concentration less than 100 mg/dL), PT, APTT, and TT are prolonged. In factor VII deficiency, APTT is normal; in factor VIII deficiency, PT is normal; and in factor X deficiency, TT is normal.
- 12. B** Coagulation factor assays are based on the ability of the patient's plasma to correct any specific factor-deficient plasma. To measure for factor VIII activity in a patient's plasma, diluted patient plasma is mixed with factor VIII-deficient plasma. The APTT test is performed on the mixture. Each laboratory should calculate its own normal ranges, which are based on the patient population, reagents, and instrument used. A range of 50% to 150% is considered normal.

13. The most suitable product for treatment of factor VIII deficiency is:

- A. Fresh frozen plasma (FFP)
- B. Factor VIII concentrate
- C. Prothrombin complex concentrate
- D. Factor V Leiden

Hemostasis/Correlate clinical and laboratory data/Treatment/2

14. Which of the following is associated with an abnormal PLT aggregation test result?

- A. Factor VIII deficiency
- B. Factor VIII inhibitor
- C. Lupus anticoagulant
- D. Afibrinogenemia

Hemostasis/Correlate clinical and laboratory data/Factor deficiency/2

15. Refer to the following results:

PT = normal

APTT = prolonged

PLT count = normal

PLT aggregation to ristocetin = abnormal

Which of the following disorders may be indicated?

- A. Factor VIII deficiency
- B. DIC
- C. von Willebrand disease
- D. Factor IX deficiency

Hemostasis/Correlate clinical and laboratory data/Coagulation disorders/3

16. Which results are associated with hemophilia A?

- A. Prolonged APTT, normal PT
- B. Prolonged PT and APTT
- C. Prolonged PT, normal APTT
- D. Normal PT and APTT

Hemostasis/Correlate clinical and laboratory data/Hemophilia/2

17. Fibrin monomers are increased in which of the following conditions?

- A. Primary fibrinolysis
- B. DIC

- C. Factor VIII deficiency
- D. Fibrinogen deficiency

Hemostasis/Correlate clinical and laboratory data/2

18. Which of the following is associated with multiple factor deficiencies?

- A. An inherited disorder of coagulation
- B. Severe liver disease
- C. Dysfibrinogenemia
- D. Lupus anticoagulant

Hemostasis/Correlate clinical and laboratory data/Factor deficiency/2

19. Normal PT and APTT results in a patient with poor wound healing may be associated with:

- A. Factor VII deficiency
- B. Factor VIII deficiency
- C. Factor XII deficiency
- D. Factor XIII deficiency

Hemostasis/Correlate clinical and laboratory data/Factor deficiency/2

Answers to Questions 13–19

- 13. B** Factor VIII concentrate (human or recombinant) is the treatment of choice for patients with factor VIII deficiency. FFP contains factor VIII; however, it is no longer used as the primary treatment for factor VIII deficiency. Prothrombin complex concentrate is used to treat patients with factor VIII inhibitor.
- 14. D** Fibrinogen is a plasma protein that is essential for PLT aggregation as well as fibrin formation. In afibrinogenemia, PLT aggregation is abnormal.
- 15. C** VWF is involved in both PLT adhesion and coagulation via complexing with factor VIII. Therefore, in von Willebrand disease (deficiency or functional abnormality of VWF) factor VIII is also decreased, causing abnormal APTT as well as abnormal PLT aggregation to ristocetin. The PLT count and PT are not affected in VWF deficiency.
- 16. A** Hemophilia A is associated with factor VIII deficiency. Factor VIII is a factor in the intrinsic coagulation pathway that is evaluated by the APTT test and not the PT test. The PT test evaluates the extrinsic and common pathways.
- 17. B** Increased fibrin monomers result from coagulation activation. DIC is an acquired condition associated with spontaneous activation of coagulation and fibrinolysis. In primary fibrinolysis, the fibrinolytic system is activated, and fibrin monomers are normal.
- 18. B** Most of the clotting factors are made in the liver. Therefore, severe liver disease results in multiple-factor deficiencies. An inherited disorder of coagulation is commonly associated with a single-factor deficiency. Lupus anticoagulant is directed against the phospholipid-dependent coagulation factors. Dysfibrinogenemia results from an abnormal fibrinogen molecule.
- 19. D** Factor XIII deficiency can lead to impaired wound healing and may cause severe bleeding problems. Factor XIII is a fibrin stabilizing factor that changes the fibrinogen

bonds in fibrin polymers to stable covalent bonds. Factor XIII is not involved in the process of fibrin formation and, therefore, PT and APTT are both normal.

20. Fletcher factor (prekallikrein) deficiency may be associated with:

- A. Bleeding
- B. Thrombosis
- C. Thrombocytopenia
- D. Thrombocytosis

Hemostasis/Correlate clinical and laboratory data/Factor deficiency/2

21. One of the complications associated with a severe hemophilia A is:

- A. Hemarthrosis
- B. Mucous membrane bleeding
- C. Mild bleeding during surgery
- D. Immune-mediated thrombocytopenia

Hemostasis/Apply knowledge of fundamental biological characteristics/Hemophilia/1

22. The most common subtype of classic von Willebrand disease is:

- A. Type 1
- B. Type 2A
- C. Type 2B
- D. Type 3

Hemostasis/Apply knowledge of fundamental biological characteristics/von Willebrand disease/2

23. Prolonged APTT and PT are corrected when mixed with normal plasma. Which factor is most likely deficient?

- A. Factor VIII
- B. Factor V
- C. Factor XI
- D. Factor IX

Hemostasis/Evaluate laboratory data to recognize health and disease states/Factor deficiency/3

Answers to Questions 20–23

- 20. B** Fletcher factor (prekallikrein) is referred to as a *contact factor*. Activated prekallikrein is named *kallikrein*. In vitro, kallikrein is involved in activation of factor XI to XIa causing prolonged APTT. In vivo, it plays a role in the fibrinolytic system and, similar to factor XII deficiency, Fletcher factor deficiency may be associated with thrombosis.
- 21. A** In severe hemophilia A, factor VIII activity is less than 1%, resulting in a severe bleeding diathesis, such as hemarthrosis (bleeding into the joints).
- 22. A** VWF is a multimeric plasma glycoprotein that results in different subtypes of von Willebrand disease with varied severity. The most common subtype is subtype 1, and 70% to 80% of these cases are associated with mild bleeding. Subtype 3 involves the total absence of the von Willebrand molecule and is associated with severe bleeding. Subtypes 2A and 2B result in deficiency of intermediate- and/or high-molecular-weight portions of the von Willebrand molecule and are associated with 10% to 12%

and 3% to 6% of cases of von Willebrand disease, respectively.

- 23. B** Factor V (common pathway factor) deficiency is most likely suspected because both PT and APTT are prolonged, and both are corrected when mixed with normal plasma.

2.4

Inhibitors, Thrombotic Disorders, and Anticoagulant Drugs

1. Which characteristic describes antithrombin (AT)?

- A. It is synthesized in megakaryocytes
- B. It is activated by protein C
- C. It is a cofactor of heparin
- D. It is a pathological inhibitor of coagulation

Hemostasis/Apply knowledge of fundamental biological characteristics/AT/1

2. Which laboratory test is affected by heparin therapy?

- A. Thrombin time
- B. Fibrinogen assay
- C. Protein C assay
- D. Protein S assay

Hemostasis/Apply knowledge of fundamental biological characteristics/Hemostasis/Heparin/2

3. Abnormal APTT caused by a pathological circulating anticoagulant is:

- A. Corrected with factor VIII-deficient plasma
- B. Corrected with factor IX-deficient plasma
- C. Corrected with normal plasma
- D. Not corrected with normal plasma

Hemostasis/Correlate clinical and laboratory data/Special tests/2

4. The lupus anticoagulant affects which of the following tests?

- A. Factor VIII assay
- B. Factor IX assay
- C. VWF assay
- D. Phospholipid-dependent assays

Hemostasis/Apply knowledge of fundamental biological characteristics/Lupus anticoagulant/2

5. Which statement about warfarin (Coumadin) is accurate?

- A. It is a vitamin B antagonist
- B. It is not recommended for pregnant and lactating women
- C. It needs AT as a cofactor
- D. APTT test is used to monitor its dosage

Hemostasis/Apply knowledge of fundamental biological characteristics/Warfarin/2

Answers to Questions 1–5

1. C AT is a heparin cofactor and is the most important naturally occurring physiological

inhibitor of blood coagulation. It represents about 75% of antithrombotic activity and is an α_2 -globulin made by the liver.

2. **A** Heparin is an AT drug and therefore increases TT along with APTT and PT. Heparin therapy has no effect on fibrinogen, protein C, or protein S assays.
3. **D** In the presence of a pathological circulating anticoagulant, a mixing test using normal plasma does not correct abnormal APTT. These anticoagulants are pathological substances and are endogenously produced. They are either directed against a specific clotting factor or against a group of factors. Prolonged APTT caused by a factor deficiency is corrected when mixed with normal plasma. Factors VIII– and factor IX–deficient plasmas are used for assaying the activities of factors VIII and IX, respectively.
4. **D** The lupus anticoagulant interferes with phospholipid-dependent coagulation assays, such as the PT and APTT tests. The lupus anticoagulant does not inhibit clotting factor assays and does not inhibit in vivo coagulation.
5. **B** Warfarin (Coumadin) crosses the placenta and is present in human milk; it is not recommended for pregnant and lactating women. Warfarin is a vitamin K antagonist drug that retards synthesis of the active form of vitamin K–dependent factors (II, VII, IX, and X). AT is a heparin (not warfarin) cofactor. The INR is used to monitor warfarin dosage.

6. Which statement regarding protein C is correct?

- A.** It is a vitamin K–independent zymogen
- B.** It is activated by fibrinogen
- C.** It activates cofactors V and VIII
- D.** Its activity is enhanced by protein S

Hemostasis/Apply knowledge of fundamental biological characteristics/Protein C/1

7. Which of the following is an appropriate screening test for the diagnosis of lupus anticoagulant?

- A.** Thrombin time
- B.** Diluted Russell viper venom test (DRVVT)
- C.** D-dimer test
- D.** Fibrinogen assay

Hemostasis/Correlate clinical and laboratory data/Lupus anticoagulant/2

8. Which of the following is most commonly associated with activated protein C resistance (APCR)?

- A.** Bleeding
- B.** Thrombosis
- C.** Epistaxis
- D.** Menorrhagia

Hemostasis/Correlate clinical and laboratory data/APCR/2

9. A 50-year-old man has been on heparin for the past 7 days. Which combination of tests is expected to be abnormal?

- A.** PT and APTT only
- B.** APTT, TT only

C. APTT, TT, fibrinogen assay

D. PT, APTT, TT

Hemostasis/Correlate clinical and laboratory data/Heparin therapy/3

10. Which of the following drugs inhibits ADP-mediated PLT aggregation?

A. Heparin

B. Warfarin

C. Aspirin

D. Prasugrel

Hemostasis/Correlate clinical and laboratory data/Therapies/2

11. Thrombin–TM complex is necessary for activation of:

A. Protein C

B. AT

C. Protein S

D. Factors V and VIII

Hemostasis/Apply knowledge of fundamental biological characteristics/Thrombomodulin/2

Answers to Questions 6–11

6. D Protein S functions as a cofactor of protein C and, as such, enhances its activity.

Activated protein C (APC) inactivates factors Va and VIIIa.

7. B Russell viper venom (RVV) reagent contains factors X and V, activating enzymes that are strongly phospholipid dependent. The reagent also contains RVV, Ca^{2+} , and phospholipid. In the presence of phospholipid autoantibodies, such as lupus anticoagulant, the reagent phospholipid is partially neutralized, causing prolongation of clotting time. TT evaluates fibrinogen. D-dimer tests evaluate fibrin degradation products. Fibrinogen assay and TT are not indicated for the diagnosis of lupus anticoagulant.

8. B APCR is the single most common cause of inherited thrombosis. In 90% of individuals, the cause is gene mutation of factor V (factor V Leiden). Affected individuals are predisposed to thrombosis, mainly after age 40 years. Heterozygous individuals may not manifest thrombosis unless other clinical conditions coexist.

9. D Heparin is a therapeutic anticoagulant with an AT activity. Heparin also inhibits factors XIIa, XIa, Xa, and IXa. In patients receiving heparin therapy, PT, APTT, and TT are all prolonged. Quantitative fibrinogen assay, however, is not affected by heparin therapy.

10. D Prasugrel (Effient) is an antiplatelet drug that reduces PLT aggregation by irreversibly blocking P2Y₁₂ receptors on the PLT surface membrane, thereby inhibiting PLT aggregation to ADP. Aspirin is another antiplatelet drug that inhibits PLT aggregation by blocking the action of the enzyme cyclo-oxygenase. Warfarin and heparin are anticoagulant drugs that act against clotting factors.

11. A Protein C is activated by thrombin–TM complex. TM is a transmembrane protein that accelerates protein C activation 1,000-fold by forming a complex with thrombin. When thrombin binds to TM, it loses its clotting function, including activation of factors V and

VIII. APC deactivates factors Va and VIIIa. Protein S is a cofactor necessary for the activation of protein C.

12. Which test is used to monitor heparin therapy?

- A. INR
- B. Chromogenic anti-factor Xa assay
- C. TT
- D. PT

Hemostasis/Correlate clinical and laboratory data/Heparin therapy/2

13. Which test is commonly used to monitor warfarin therapy?

- A. INR
- B. APTT
- C. TT
- D. Ecarin time

Hemostasis/Correlate clinical and laboratory data/Warfarin therapy/2

14. Which clotting factors (cofactors) are inhibited by protein S?

- A. Factors V and X
- B. Factors Va and VIIIa
- C. Factors VIII and IX
- D. Factors VIII and X

Hemostasis/Correlate clinical and laboratory data/Clotting factors/2

15. Which drug promotes fibrinolysis?

- A. Warfarin
- B. Heparin
- C. Urokinase
- D. Aspirin

Hemostasis/Correlate clinical and laboratory data/Therapies/2

16. Diagnosis of lupus anticoagulant is confirmed by which of the following criteria?

- A. Decreased APTT
- B. Correction of APTT by mixing studies
- C. Neutralization of the antibody by high concentration of phospholipids
- D. Confirmation that abnormal coagulation tests are related to factor deficiencies

Hemostasis/Correlate clinical and laboratory data/LA/3

17. Which of the following abnormalities is consistent with the presence of lupus anticoagulant?

- A. Decreased APTT/bleeding complications
- B. Prolonged APTT/thrombosis
- C. Prolonged APTT/thrombocytosis
- D. Thrombocytosis/thrombosis

Hemostasis/Correlate clinical and laboratory data/LA/3

Answers to Questions 12–17

- 12. B** Heparin dosage can be best monitored by the chromogenic anti-factor Xa assay. In anti-factor Xa assay, the concentration of heparin is determined by inhibition of factor Xa by AT. Anti-factor Xa assay uses a reagent with a fixed concentration of factor Xa

and AT. Heparin forms a complex with AT and factor Xa reagents. Excess factor Xa combines with the chromogenic substrate to form a colored product; the color intensity is inversely proportional to the concentration of heparin. PT would be prolonged in heparin therapy, but the PT test is not sensitive enough to be used to monitor heparin therapy. Heparin inhibits thrombin and, therefore, causes prolonged TT. The TT test, however, is not used to monitor heparin therapy either.

- 13. A** Warfarin is a vitamin K–antagonist drug. It inhibits vitamin K–dependent factors (II, VII, IX, and X) and other vitamin K–dependent proteins, such as proteins C and S. Warfarin therapy is monitored with the INR. An INR of 2.0 to 3.0 is used as the target when monitoring warfarin therapy for prophylaxis and treatment of deep vein thrombosis (DVT). A higher dose of warfarin (giving an INR of 2.5–3.5) is required for patients with mechanical heart valves.
- 14. B** Factors Va and VIIIa are deactivated by protein S and activated by protein C.
- 15. C** Urokinase is a thrombolytic drug that can be used to treat acute arterial thrombosis. Urokinase can also be used for the treatment of venous thromboembolism, myocardial infarction, and clotted catheters. Warfarin and heparin are anticoagulant drugs, whereas aspirin prevents PLT aggregation by inhibiting cyclo-oxygenase.
- 16. C** The International Society of Hemostasis and Thrombosis has recommended four criteria for the diagnosis of lupus anticoagulant: (1) prolongation of one or more of the phospholipid-dependent clotting tests, such as APTT or DRVVT; (2) presence of an inhibitor confirmed by mixing studies (not corrected); (3) evidence that the inhibitor is directed against phospholipids by neutralizing the antibodies with a high concentration of phospholipids (PLT neutralization test or DRVVT with platelet rich plasma or confirming phospholipid reagent); and (4) lack of any other causes for thrombosis.
- 17. B** Lupus anticoagulant interferes with phospholipids in the APTT reagent, resulting in prolongation of APTT. However, in vivo, lupus anticoagulant decreases fibrinolytic activity, causing an increased risk of thrombosis. Lupus anticoagulant does not result in a bleeding tendency unless there is a coexisting thrombocytopenia or other coagulation abnormality.

18. Which of the following is a characteristic of LMWH?

- A. Generally requires monitoring
- B. Specifically acts on factor Va
- C. Has a longer half-life compared with unfractionated heparin (UFH)
- D. Can be used as a fibrinolytic agent

Hemostasis/Apply knowledge of fundamental biological characteristics/LMWH/1

19. Which of the following tests is most likely to be abnormal in patients taking aspirin?

- A. PLT morphology
- B. PLT count
- C. PLT aggregation
- D. PT

Hemostasis/Correlate clinical and laboratory data/Aspirin therapy/2

20. Which of the following is associated with AT deficiency?

- A. Thrombocytosis

- B. Thrombosis
- C. Thrombocytopenia
- D. Bleeding

Hemostasis/Correlate clinical and laboratory data/Inhibitors/2

21. Which of the following may be associated with thrombotic events?

- A. Decreased protein C
- B. Increased fibrinolysis
- C. Afibrinogenemia
- D. Idiopathic thrombocytopenic purpura

Hemostasis/Correlate clinical and laboratory data/Protein C/2

22. Aspirin resistance may be associated with:

- A. Bleeding
- B. Factor VIII deficiency
- C. Thrombosis
- D. Thrombocytosis

Hemostasis/Correlate clinical and laboratory data/Aspirin resistance/2

23. Prolonged TT is indicative of which of the following antithrombotic agents?

- A. Prasugrel
- B. Clopidogrel
- C. Aspirin
- D. Heparin

Hemostasis/Correlate clinical and laboratory data/Antithrombotic therapy/2

24. Screening tests for thrombophilia should be performed on:

- A. All pregnant women because of the risk of thrombosis
- B. Patients with a negative family history
- C. Patients with thrombotic events occurring at a young age
- D. Patients who are receiving anticoagulant therapy

Hemostasis/Correlate clinical and laboratory data/Thrombophilia/2

Answers to Questions 18–24

- 18. C** LMWH is a small glycosaminoglycan that is derived from UFH. LMWH has a low affinity for plasma proteins and endothelial cells and therefore has a longer half-life. The half-life of the drug does not depend on the dosage. LMWH has an inhibitory effect on factors Xa and IIa. It does not require routine monitoring except in patients with renal failure, obese patients, pediatric patients, and pregnant patients.
- 19. C** Aspirin is an antiplatelet drug. It prevents PLT aggregation by inhibition of cyclo-oxygenase. Aspirin has no effect on the PLT count, PLT morphology, or PT.
- 20. B** AT is a physiological anticoagulant. It inhibits factors IIa, Xa, IXa, XIa, and XIIa. Deficiency of AT is associated with thrombosis. Thrombotic events commonly occur when AT deficiency is associated with another risk factor, such as pregnancy, surgery, or an inherited thrombotic disorder (e.g., factor V Leiden).
- 21. A** Protein C is a physiological inhibitor of coagulation. It is activated by thrombin–TM complex. APC inhibits cofactors Va and VIIIa. Protein C deficiency is associated with

thrombosis. Increased fibrinolysis, afibrinogenemia, and ITP are associated with bleeding.

22. **C** Up to 22% of patients taking aspirin become resistant to aspirin's antiplatelet effect. Patients who are aspirin resistant have a higher risk of thrombosis (heart attacks and strokes).
23. **D** Heparin is an AT drug causing prolonged TT in patients who are on heparin therapy. Prasugrel, clopidogrel, and aspirin are antiplatelet drugs causing inhibition of PLT aggregation.
24. **C** Laboratory tests for evaluation of thrombophilia are justified in young patients with thrombotic events, in patients with a positive family history after a single thrombotic event, in those with recurrent spontaneous thrombosis, and in pregnancies associated with thrombosis.

25. Prothrombin G20210A is characterized by which of the following causes and conditions?

- A. Single mutation of prothrombin molecule/bleeding
- B. Single mutation of prothrombin molecule/thrombosis
- C. Decreased levels of prothrombin in plasma/thrombosis
- D. Increased levels of prothrombin in plasma/bleeding

Hemostasis/Correlate clinical and laboratory data/Prothrombin/2

26. Factor V Leiden promotes thrombosis by preventing:

- A. Inactivation of factor Va
- B. Activation of factor V
- C. Activation of protein C
- D. Activation of protein S

Hemostasis/Correlate clinical and laboratory data/Factor V Leiden/2

27. What is the approximate incidence of antiphospholipid antibodies in the general population?

- A. Less than 1%
- B. 1%–2%
- C. 3%–8%
- D. 10%–15%

Hemostasis/Apply knowledge of fundamental biological characteristics/LA/1

28. Which of the following laboratory tests is helpful in the diagnosis of aspirin resistance?

- A. APTT
- B. PT
- C. PLT count and morphology
- D. PLT aggregation

Hemostasis/Correlate clinical and laboratory data/Aspirin resistance/2

29. Which of the following complications may occur as a result of decreased tissue factor pathway inhibitor (TFPI)?

- A. Increased episodes of hemorrhage
- B. Increased risk of thrombosis
- C. Impaired PLT plug formation

D. Immune thrombocytopenia

Hemostasis/Apply knowledge of fundamental biological characteristics/TFPI/2

30. Factor VIII inhibitors occur in _____ of patients with factor VIII deficiency.

A. 40%–50%

B. 30%–40%

C. 25%–30%

D. 20%–25%

Hemostasis/Apply knowledge of fundamental biological characteristics/Inhibitors/1

Answers to Questions 25–30

25. B Prothrombin G20210A is defined as a single-point mutation of the prothrombin gene, resulting in increased concentration of plasma prothrombin and, thus, is a risk factor for thrombosis. Prothrombin G20210A is the second most common cause of inherited hypercoagulability (after factor V Leiden). It has the highest incidence in white people from southern Europe. The thrombotic episodes generally occur before age 40 years.

26. A Factor V Leiden is a single-point mutation in the factor V gene that inhibits factor Va inactivation by protein C. APC enhances deactivation of factors Va and VIIa.

27. B The incidence of antiphospholipid antibodies in the general population is about 1% to 2%.

28. D Currently, the PLT aggregation test is considered the gold standard for evaluation of aspirin resistance. In aspirin resistance, PLT aggregation is not inhibited by aspirin ingestion. Aspirin resistance has no effect on PLT count and morphology.

29. B TFPI is released from the vasculature and is the most important inhibitor of the extrinsic pathway. TFPI inhibits factors Xa and VIIa–TF complex. Therefore, the deficiency of TFPI is associated with thrombosis.

30. D Factor VIII inhibitors (alloantibodies) occur in 20% to 25% of patients who have factor VIII deficiency and are receiving factor VIII replacement.

31. Which therapy and resulting mode of action are appropriate for the treatment of a patient with a high titer of factor VIII inhibitors?

A. Factor VIII concentrate to neutralize the antibodies

B. Recombinant factor VIIa (rVIIa) to activate factor X

C. Factor X concentrate to activate the common pathway

D. FFP to replace factor VIII

Hemostasis/Apply knowledge of fundamental biological characteristics/Inhibitors/2

32. The Bethesda assay is used for which determination?

A. Lupus anticoagulant titer

B. Factor VIII inhibitor titer

C. Factor V Leiden titer

D. Protein S deficiency

Hemostasis/Select methods/Reagents/Special tests/2

33. Hyperhomocysteinemia may be a risk factor for:

A. Bleeding

- B. Thrombocythemia
- C. Thrombosis
- D. Thrombocytopenia

Hemostasis/Correlate clinical and laboratory data/Homocysteinemia/2

34. Which drug may be associated with DVT?

- A. Aspirin
- B. tPA
- C. Oral contraceptives
- D. Clopidogrel (Plavix)

Hemostasis/Apply knowledge of fundamental biological characteristics/Thrombosis/2

35. Argatroban may be used as an anticoagulant drug in patients with:

- A. DVT
- B. Hemorrhage
- C. TTP
- D. Thrombocytosis

Hemostasis/Apply knowledge of fundamental biological characteristics/Therapies/2

36. Heparin-induced thrombocytopenia (HIT) results from:

- A. Antibodies to heparin
- B. Antibodies to PLTs
- C. Antibodies to PF4
- D. Antibodies to heparin–PF4 complex

Hemostasis/Apply knowledge of fundamental biological characteristics/HIT/2

Answers to Questions 31–36

- 31. B** rVIIa is effective for the treatment of a high-titer factor VIII inhibitor. Factor VIIa can directly activate factor X to factor Xa in the absence of factors VIII and IX. rVIIa does not stimulate anamnestic responses in patients with factor VIII inhibitor. Factor VIII concentrate is used for a low-titer factor VIII inhibitor. Factor X concentrate and FFP are not the treatments of choice for factor VIII inhibitor.
- 32. B** The Bethesda assay is a quantitative assay for factor VIII inhibitor. In this assay, normal plasma is incubated with different dilutions of the patient's plasma or with a normal control. The inhibitor inactivates factor VIII present in normal plasma following incubation for 2 hours at 37°C. The residual activities in the sample are determined, and the inhibitor titer is calculated.
- 33. C** Elevated plasma homocysteine is a risk factor for the development of DVT, coronary heart disease and stroke. Homocystinemia may be inherited or acquired. Acquired homocystinemia is caused by dietary deficiencies of vitamins B₆, B₁₂, and folic acid.
- 34. C** Oral contraceptive drugs are acquired risk factors for thrombosis. Aspirin and clopidogrel are antiplatelet drugs, and tPA is a fibrinolytic drug used for the treatment of thrombosis.
- 35. A** Argatroban is a direct thrombin-inhibiting drug and may be used as an anticoagulant in patients with HIT to prevent thrombosis. Argatroban is a small synthetic molecule that binds to free and clot-bound thrombin. Argatroban affects TT, PT, APTT, and

activated clotting time (ACT) tests. The APTT test is recommended for monitoring the dosage with the target therapeutic range of 1.5 to 3.0 times the mean of the laboratory reference range. In patients with lupus anticoagulant or factor deficiencies, baseline APTT is prolonged; in these conditions, Ecarin time can be used as an alternative assay.

- 36. D** HIT is an immune process caused by the production of antibodies to heparin–PF4 complex. This immune complex binds to PLT Fc receptors, causing PLT activation and formation of PLT microparticles, which, in turn, induce hypercoagulability and thrombocytopenia.

37. Which laboratory test is used to screen for APCR?

- A. Mixing studies with normal plasma
- B. Mixing studies with factor-deficient plasma
- C. Modified APTT with and without APC
- D. Modified PT with and without APC

Hemostasis/Select methods/Reagents/Special tests/2

38. Ecarin clotting time may be used to monitor:

- A. Heparin therapy
- B. Warfarin therapy
- C. Fibrinolytic therapy
- D. Bivalirudin

Hemostasis/Select methods/Reagents/Special tests/2

39. Which of the following may interfere with the APCR screening test?

- A. Lupus anticoagulant
- B. Protein C deficiency
- C. AT deficiency
- D. Protein S deficiency

Hemostasis/Correlate clinical and laboratory/Special tests/2

40. Thrombophilia may be associated with which of the following disorders?

- A. Afibrinogenemia
- B. Hypofibrinogenemia
- C. Factor VIII inhibitor
- D. Hyperfibrinogenemia

Hemostasis/Apply knowledge of fundamental biological characteristics/Fibrinogen/2

41. Which of the following anticoagulant drugs can be used in patients with HIT?

- A. Warfarin
- B. Heparin
- C. Aspirin
- D. Argatroban

Hemostasis/Apply knowledge of fundamental biological characteristics/Therapies/2

Answers to Questions 37–41

- 37. C** APCR can be evaluated by performing a two-part APTT test. The APTT is measured on the patient's plasma with and without the addition of APC. The result is expressed

as the ratio of the APTT with APC to the APTT without APC. The normal ratio is 2:5. Patients with APCR have a lower ratio than the reference range. A positive screening test should be followed by a confirmatory test, such as polymerase chain reaction (PCR) for factor V Leiden.

38. **D** Ecarin clotting time, a snake venom–based clotting assay, may be used to monitor bivalirudin therapy in instances when the baseline APTT is prolonged as a result of lupus anticoagulant or factor deficiencies. Heparin therapy is monitored by using the anti–factor Xa assay; warfarin therapy is monitored with the INR. Fibrinolytic therapy may be monitored by using the D-dimer test.
39. **A** The lupus anticoagulant interferes with the APCR screening assay based on the APTT ratio with and without the addition of APC. Persons with the lupus anticoagulant have a prolonged APTT that renders the test invalid for APCR screening.
40. **D** Hyperfibrinogenemia is a risk factor for thrombophilia. Fibrinogen is an acute-phase reactant and may be increased in inflammation, stress, obesity, smoking, and medications, such as oral contraceptives. Hypofibrinogenemia, afibrinogenemia, and factor VIII inhibitors are associated with bleeding.
41. **D** Argatroban is a direct thrombin inhibitor drug and used in patients with HIT who cannot tolerate heparin or LMWH therapy. Warfarin should not be used for anticoagulation in persons with HIT because it causes a fall in protein C concentration and vitamin K–dependent coagulation factors. Aspirin is an antiplatelet drug.

42. Which of the following is the preferred method to monitor heparin therapy at the point of care during cardiac surgery?

- A. APTT
- B. ACT
- C. PT
- D. TT

Hemostasis/Correlate clinical and laboratory/Special tests/2

43. Mrs. Smith has the following laboratory results and no history of bleeding:

APTT = prolonged

APTT results on a 1:1 mixture of the patient's plasma with normal plasma:

Preincubation: prolonged APTT

2-hour incubation: prolonged APTT

These results are consistent with:

- A. Factor VIII deficiency
- B. Factor VIII inhibitor
- C. Lupus anticoagulant
- D. Protein C deficiency

Hemostasis/Correlate clinical and laboratory data/Special tests/3

44. Which test may be used to monitor LMWH therapy?

- A. APTT
- B. INR

- C. Anti-factor Xa heparin assay
- D. ACT

Hemostasis/Correlate clinical and laboratory data/LMWH therapy/2

Answers to Questions 42–44

- 42. **B** ACT is a point-of-care coagulation test used to monitor high-dose heparin therapy during cardiac surgery, cardiac angioplasty, hemodialysis, and other major surgeries. It is the preferred method to determine if sufficient heparin was administered to prevent clotting during surgery because it is more rapid than the APTT test. The test uses a clot activator, such as kaolin or Celite, to stimulate coagulation, and the time in seconds is linearly related to the dose of heparin administered. The ACT test is available in different formats, and the reference range varies, depending on the method used. At low to moderate heparin doses, the ACT test does not correlate well with the APTT test or the anti-factor Xa assay.
- 43. **C** Mixing studies differentiate factor deficiencies from factor inhibitors. Lupus anticoagulant is associated with thrombosis, and it is directed against phospholipid-dependent coagulation tests, such as the APTT test. In patients with lupus anticoagulant, after the patient's plasma is mixed with normal plasma, APTT remains prolonged immediately after mixing and following 2 hours of incubation. Factor VIII deficiency and factor VIII inhibitor are associated with bleeding.
- 44. **C** The anti-factor Xa heparin assay is used to monitor LMWH therapy, when required, because the APTT test is insensitive to LMWH. The assay can be performed by chromogenic endpoint detection used on automated analyzers. The principle of the test is to measure the inhibition of factor Xa by heparin. The reagent is a mixture of a fixed concentration of factor Xa, a substrate which is specific for factor Xa, and a fixed concentration of AT. Some kits rely on the AT in the patient's plasma. Heparin forms a complex with AT and factor Xa (AT-heparin-factor Xa). Excess free factor Xa cleaves the chromogenic substrate and releases a yellow product. The color intensity of the product is inversely proportional to plasma heparin concentration and is measured by a photodetector at 405 nm. LMWH therapy usually does not require monitoring; however, exceptions include pediatric, obese, and pregnant patients and those with renal failure.

2.5

Hemostasis Problem-Solving

1. Patient History

A 3-year-old male was admitted with scattered petechiae and epistaxis. The patient had normal growth and had no other medical problems except for chickenpox 3 weeks earlier. His family history was unremarkable.

Laboratory Results		
	Patient	Reference Range
PT	11 sec	10–13 sec
APTT	32 sec	28–37 sec
PLT count	$18 \times 10^3/\mu\text{L}$	$150\text{--}450 \times 10^3/\mu\text{L}$

These clinical manifestations and laboratory results are consistent with which condition?

- A. TTP
- B. DIC
- C. ITP
- D. HUS

Hemostasis/Evaluate laboratory data to recognize health and disease states/Platelet disorders/3

Answer to Question 1

1. **C** These clinical manifestations and laboratory results are consistent with ITP. ITP is an autoimmune thrombocytopenia. In children, acute ITP occurs after a viral infection, as was the case in this 3-year-old patient. Clinical manifestations are associated with petechiae, purpura, and mucous membrane bleeding, such as epistaxis and gingival bleeding. Abnormal laboratory tests include a very low PLT count, and other causes of thrombocytopenia should be ruled out in patients with suspected ITP.

2. Patient History

A 12-year-old white male has the following symptoms: visible bruising on arms and legs, bruising after sports activities, and excessive postoperative hemorrhage after tonsillectomy 3 months ago. His family history revealed that his mother suffers from heavy menstrual bleeding, and his maternal grandfather had recurrent nosebleeds and bruising.

Laboratory Results		
	Patient	Reference Range

PLT count:	$350 \times 10^3/\mu\text{L}$	$200\text{--}450 \times 10^3/\mu\text{L}$
PT	11 sec	10–12 sec
APTT	70 sec	28–37 sec
TT	13 sec	10–15 sec
PLT AGGREGATION		
Normal aggregation with collagen, EPI, ADP		
Abnormal aggregation with ristocetin		
CONFIRMATORY TESTS	PATIENT	REFERENCE RANGE
VWF:Rco	25%	45%–140%
VIII:C	20%	50%–150%
VWF:antigen	10%	45%–185%

These clinical manifestations and laboratory results are consistent with which diagnosis?

- A. Factor VIII deficiency
- B. von Willebrand disease
- C. Glanzmann thrombasthenia
- D. Bernard-Soulier syndrome

Hemostasis/Evaluate laboratory data to recognize health and disease states/Platelet disorders/3

3. The following results are obtained from a patient who developed severe bleeding:

Prolonged PT and APTT

PLT count = $100 \times 10^9/\text{L}$

Fibrinogen = 40 mg/dL

Which of the following blood products should be recommended for transfusion?

- A. Factor VIII concentrate
- B. PLTs
- C. FFP
- D. Cryoprecipitate

Hemostasis/Correlate clinical and laboratory data/Therapies/2

4. A 30-year-old woman develops signs and symptoms of thrombosis in her left lower leg after 5 days of heparin therapy. The patient had had open-heart surgery 3 days previously and has been on heparin ever since. Which of the following would be most helpful in making the diagnosis?

- A. Fibrinogen assay
- B. PT
- C. PLT count
- D. Increased heparin dose

Hemostasis/Correlate clinical and laboratory data/Heparin therapy/3

Answers to Questions 2–4

2. **B** These clinical manifestations and laboratory results are consistent with von Willebrand disease. von Willebrand disease is an inherited bleeding disorder caused by abnormal PLT adhesion and aggregation. In von Willebrand disease, VWF is deficient or dysfunctional. VWF enhances the release of factor VIII from the liver, and it also forms a complex with factor VIII in the circulation. Deficient or dysfunctional VWF results in decreased factor VIII as well and, therefore, an increase in APTT values. The clinical manifestations associated with von Willebrand disease are easy bruising, epistaxis, and bleeding after surgery. The diagnostic laboratory test shows abnormal PLT aggregation to ristocetin, which is corrected by addition of normal plasma containing VWF. APTT is prolonged as a result of the deficiency of factor VIII. Factor VIII activity (VIII:C), VWF ristocetin cofactor activity (VWF:Rco), and VWF: antigenic activity (VWF:antigen) are abnormal. The PLT count and PT are normal in von Willebrand disease.
3. **D** Cryoprecipitate contains fibrinogen, factor VIII, and VWF. FFP has all of the clotting factors; however, it is not the best choice if cryoprecipitate is available.
4. **C** The PLT count should be checked every other day in patients receiving heparin therapy. HIT should be suspected in patients who are not responding to heparin therapy and/or are developing thrombocytopenia (50% below the baseline value) and thrombotic complications while on heparin therapy. Increase in heparin dose should be avoided in patients with the clinical symptoms of thrombosis while they are receiving heparin. Fibrinogen assay and PT are not the appropriate assays for monitoring heparin therapy, nor are they used to test for HIT.

5. **The following laboratory results were obtained on a 25-year-old woman with menorrhagia after delivery of her second son. The patient has no previous bleeding history.**

Normal PLT count; normal PT; prolonged APTT
Mixing of the patient's plasma with normal plasma corrected the prolonged APTT on immediate testing.
However, mixing followed by 2-hour incubation at 37°C caused prolonged APTT.

What is the most probable cause of these laboratory results?

- A. Lupus anticoagulant
- B. Factor VIII deficiency
- C. Factor IX deficiency
- D. Factor VIII inhibitor

Hemostasis/Evaluate laboratory data to recognize health and disease states/Mixing studies/3

6. **A 62-year-old female presents with jaundice and the following laboratory data:**

Peripheral blood smear = macrocytosis, target cells
PLT count = $355 \times 10^9/L$
PT = 25 sec (reference range = 10–14)
APTT = 65 sec (reference range = 28–36)

Transaminases = elevated (AST:ALT ratio greater than 1)

Total and direct bilirubin = elevated

These clinical presentations and laboratory results are consistent with:

- A. Inherited factor VII deficiency
- B. DIC
- C. Cirrhosis of the liver
- D. von Willebrand disease

Hemostasis/Correlate clinical and laboratory data/Coagulation disorders/3

7. **When performing a mixing study, the patient's APTT is corrected to 12% of normal. What is the most appropriate interpretation of these findings?**

- A. The APTT is considered corrected
- B. The APTT is considered uncorrected
- C. The laboratory protocol should be followed for the interpretation of correction
- D. A circulating anticoagulant can be ruled out

Hemostasis/Correlate clinical and laboratory data/Mixing studies/3

Answers to Questions 5–7

5. **D** Factor VIII inhibitor is found in 20% to 25% of patients with hemophilia receiving replacement therapy. It may also develop in patients with immunologic problems, women after childbirth, and patients with lymphoproliferative and plasma cell disorders, or it may develop in response to medications. Factor VIII inhibitor is an IgG with an inhibitory effect that is time and temperature dependent. The presence of factor VIII inhibitor causes elevated APTT in the face of a normal PT. Mixing studies in factors VIII and IX deficiencies will correct the prolonged APTT both at the immediate mixing stage and after incubation for 2 hours. APTT would not be corrected by mixing studies in the presence of factor VIII inhibitor. In addition, factor VIII inhibitor is associated with bleeding. Lupus anticoagulant is associated with thrombosis and not bleeding unless it coexists with thrombocytopenia, which is not the case in this patient.
6. **C** The clinical presentation and laboratory results in this patient are indicative of cirrhosis of the liver. Most of the clotting factors are made in the liver. A decrease in multiple clotting factors is associated with prolonged PT and APTT. Macrocytosis and target cells are present in liver disease. The liver changes the unconjugated bilirubin to conjugated bilirubin. Conjugated bilirubin is excreted into the intestines, where bilirubin is converted to urobilinogen and excreted into feces. In cirrhosis of the liver, both necrosis and obstruction caused by scarring produce increases in unconjugated and conjugated bilirubin, respectively. In addition, the liver enzymes are elevated (the AST:ALT ratio is less than 1 in necrotic liver diseases, such as hepatitis, but not in cirrhosis).
7. **C** Interpretation of correction studies varies among different laboratories. Some define correction when the mixing study result falls within 10% of the normal plasma; other laboratories conclude correction when the mixing result falls within 5 seconds of normal plasma, or a value within the APTT reference range. Only 50% factor activity is required for normal PT or APTT. A circulating anticoagulant typically results in failure to correct the APTT with normal plasma.

8. A standard blue-top tube filled appropriately (with 4.5 mL blood) was submitted to the laboratory for preoperative PT and APTT testing. The results of both tests were elevated. The patient's PT and APTT from the previous day were within normal limits, and he was not on heparin therapy. Which is the most appropriate first step to investigate the abnormal results?

- A. Report the result as obtained
- B. Perform a mixing study
- C. Check the sample for a clot
- D. Report APTT only

Hemostasis/Apply knowledge to identify sources of errors/Specimens/3

9. A plasma sample submitted to the laboratory for PT testing has been stored for 25 hours at 4°C. PT is shortened. What is the most probable cause?

- A. Factor VII deficiency
- B. Activation of factor VII caused by exposure to cold temperature
- C. Lupus inhibitor
- D. Factor X inhibitor

Hemostasis/Apply knowledge to identify sources of errors/Specimen storage/3

10. APTT is not increased in a patient receiving heparin. Which of the following factors may be associated with the lack of response to heparin therapy in this patient?

- A. Protein C deficiency
- B. AT deficiency
- C. Protein S deficiency
- D. Factor VIII deficiency

Hemostasis/Correlate clinical and laboratory data/Inhibitors/3

11. A 50-year-old patient was admitted to the emergency department with a complaint of pain in the right leg. The leg was red, swollen, and warm to the touch. DVT was suspected, and the patient was started on heparin therapy. Which of the following is (are) the proper protocol(s) to evaluate patients receiving heparin therapy?

- A. Baseline APTT and PLT count; APTT testing every 4 to 6 hours after the initial heparin bolus
- B. Repeat APTT 5 days after heparin therapy to adjust the therapeutic dose
- C. Monitor the PLT count daily and every other day after heparin therapy is completed
- D. Monitor PT daily to adjust the therapeutic dose

Hemostasis/Correlate clinical and laboratory data/Heparin therapy/2

Answers to Questions 8–11

- 8. C** A clot can form because of inadequate mixing of the sample after venipuncture, if the blood fills the evacuated tube at a slow rate, or with traumatic venipuncture. In vitro, blood clots result in consumption of the clotting factors and, therefore, prolongation of PT, APTT, and other clot-based assays. If the clotting factors have been activated but the clot formation is incomplete, it may result in shortening of PT and APTT. Checking the sample for a clot is the most reasonable step in this case.
- 9. B** Samples for evaluation of PT are stable for 24 hours if kept at room temperature. Prolonged exposure to cold will activate factor VII, resulting in decreased PT.

- 10. B** AT deficiency in patients receiving heparin therapy may lead to heparin resistance and, therefore, lack of prolongation of APTT. AT is a heparin cofactor and, as such, increases heparin activity by 1,000-fold. Deficiency of AT is associated with poor response to heparin therapy.
- 11. A** Baseline PLT count and APTT should be performed on all patients prior to administration of heparin. APTT should be repeated every 4 to 6 hours after bolus injection (high dose). Response to heparin therapy varies among patients for the following reasons: heparin half-life is decreased in extended thrombosis, and the anticoagulant activities of heparin change based on nonspecific binding of heparin to plasma proteins. Therefore, heparin therapy should be closely monitored. Heparin dosage can be adjusted based on the anti-factor Xa assay. In addition, the PLT count should be monitored regularly during heparin therapy because decrease of the PLT count to 50% below the baseline value is significant and may be associated with HIT. PT is not used to monitor heparin therapy.

12. Patient History:

A 46-year-old female was admitted to the emergency department with complaints of headache, dizziness, lethargy, nausea, vomiting, and weakness. The patient had undergone a gastrectomy procedure 4 months earlier for removal of adenocarcinoma of the stomach and had been placed on mitomycin therapy. Diagnostic procedures indicated recurrence of the carcinoma.

Admission Complete Blood Count (CBC) Results		
	Patient	Reference Range
WBCs	$17.1 \times 10^9/\text{L}$	$4.8\text{--}10.8 \times 10^9/\text{L}$
RBCs	$2.29 \times 10^{12}/\text{L}$	$3.80\text{--}5.50 \times 10^{12}/\text{L}$
Hgb	8.1 g/dL	12.0–15.2 g/dL
Hct	23%	37%–46%
MCV	95.7 fL	79–101 fL
MCH	35.4 pg	27–33 pg
MCHC	35.0 %	31%–34 %
RDW	18.5	11.5–14.5
PLTs	$48.0 \times 10^9/\text{L}$	$140\text{--}450 \times 10^9/\text{L}$
MPV	11.2	7.4–9.4
DIFFERENTIAL COUNTS (%)		
Segmented neutrophils	79	30%–70%
Band neutrophils	3	0%–10%
Lymphocytes	11	20%–50%
Monocytes	6	2%–12%

Basophils	1	0%–2%
NRBCs (/100 WBCs)	3	0
Manual platelet count	$18 \times 10^9/\text{L}$	$140\text{--}450 \times 10^9/\text{L}$
Marked anisocytosis		None
Marked RBC fragmentation		None
PT, APTT, and TT	Normal	
ADDITIONAL LABORATORY DATA		
Urinalysis	PATIENT	REFERENCE RANGE
pH	5.0	5–7
Protein	30.0 mg/dL	0–15 mg/dL
RBCs	60–100/ μ .	
Casts	10/high-power field (hpf) granular/hyaline	Not detectable
Plasma	PATIENT	REFERENCE RANGE
Creatinine	3.1 mg/dL	0.7–1.3 mg/dL
BUN	39 mg/dL	8–22 mg/dL
Haptoglobin	5.0 mg/dL	50–150 mg/dL

These clinical manifestations and laboratory results are consistent with:

- A. ITP
- B. von Willebrand disease
- C. TTP
- D. DIC

Hemostasis/Correlate clinical and laboratory data/Platelet disorders/3

Answer to Question 12

- 12. C** The clinical manifestations and laboratory results in this patient are consistent with TTP. The clinical manifestations of TTP include MAHA, thrombocytopenia, fever, renal failure, and neurological symptoms. The neurological symptoms in this patient are manifested by headache, dizziness, nausea, and vomiting. Weakness and lethargy are signs and symptoms of anemia. Low Hgb and Hct with normal MCV and MCHC indicate a normocytic normochromic anemia. The presence of schistocytes in peripheral blood, with low platelet counts and low haptoglobin, are consistent with MAHA. The high BUN and creatinine levels are characteristic of renal failure. The platelet count, performed on admission, was done on a hematology analyzer and was

falsely elevated because of the presence of microcytes or fragmented RBCs. The manual platelet count was much lower. The coagulation tests are normal in TTP. In von Willebrand disease, the platelet count is normal and the APTT is usually abnormal. ITP is characterized by thrombocytopenia but not hemolytic anemia (HA). DIC is associated with a low platelet count, HA, and abnormal coagulation studies. The acute onset of symptoms in this patient may be related to mitomycin used for the treatment of gastric carcinoma in this patient.

13. Patient History

A 1-year-old infant was admitted with recurrent epistaxis for the past 5 days. Past medical history revealed easy bruising and a severe nosebleed that had occurred when he was 3 months of age, necessitating transfusion therapy. The mother had a severe nosebleed 8 years ago. The father was reported to bleed easily after lacerations. The patient was transfused with 2 units of packed RBCs on admission.

Admission Laboratory Results		
	Patient	Reference Range
Hgb	4.5 g/dL	13–15 g/dL
Platelet count	$249 \times 10^9/L$	$150\text{--}450 \times 10^9/L$
PT	11.2 sec	11–13 sec
APTT	34 sec	28–37 sec
ADDITIONAL LABORATORY TESTS		
Factor VIII assay	70%	50%–150%
PLT aggregation:	Abnormal to ADP, EPI, and thrombin; normal to ristocetin	

These clinical manifestations and laboratory results are consistent with which condition?

- A.** von Willebrand disease
- B.** Bernard-Soulier syndrome
- C.** Glanzmann thrombasthenia
- D.** Factor VIII deficiency

Hemostasis/Correlate clinical and laboratory data/Platelet disorders/3

Answer to Question 13

- 13. C** These clinical manifestations and laboratory results are consistent with Glanzmann thrombasthenia. Epistaxis and easy bruising are characteristics of platelet disorders. The positive family history is indicative of an inherited bleeding disorder. Laboratory tests reveal a low Hgb level caused by epistaxis. The normal platelet count rules out any quantitative platelet disorder. The platelet count is typically low in Bernard–Soulier syndrome. Normal PT and APTT, combined with a normal factor VIII assay, rule out coagulation disorders. The laboratory tests that confirm an inherited platelet disorder are PLT aggregation studies. PLT aggregation is normal to ristocetin and

abnormal to ADP, EPI, and thrombin. These results are consistent with Glanzmann thrombasthenia. PLT aggregation is abnormal to ristocetin in von Willebrand disease and Bernard–Soulier syndrome.

14. Patient History:

A 30-year-old female was referred to the hospital for evaluation for multiple spontaneous abortions and current complaint of pain and swelling in her right leg. Her family history is unremarkable.

Laboratory Tests	Patient	Reference Range
PT	14.5 sec	11–13 sec
APTT	63.0 sec	28–37 sec
TT	12.0 sec	10–15 sec
Mixing Study APTT		
Preincubation and after 2-hour incubation at 37°C	57.0 sec	
Platelet neutralization procedure		
Patient plasma + freeze-thawed platelets	APTT = 35 sec	
Patient plasma + saline	APTT = 59 sec	
Anticardiolipin antibodies done by ELISA	Negative	

These clinical manifestations and laboratory results are consistent with:

- A.** Factor VIII inhibitor
- B.** Factor VIII deficiency
- C.** Anticardiolipin antibodies
- D.** Lupus anticoagulant

Hemostasis/Correlate clinical and laboratory data/Inhibitors/3

15. A 60-year-old patient was admitted to a hospital for a liver biopsy. The biopsy was scheduled for 11:00 a.m. The coagulation results obtained at the time of admission revealed prolonged PT with an INR of 4.5. What is the physician’s most appropriate course of action?

- A.** Proceed with biopsy because prolonged PT is expected in liver disease
- B.** Postpone the procedure for a couple of days
- C.** Cancel the procedure and start the patient on vitamin K therapy
- D.** Put patient on vitamin K therapy and proceed with the procedure immediately

Hemostasis/Correlate clinical and laboratory data/INR/3

Answers to Questions 14–15

- 14. D** These clinical manifestations and laboratory results are consistent with lupus anticoagulant. Pain and swelling in the patient’s right leg may be indicative of thrombosis. As many as 48% of women with repeated spontaneous abortions have lupus anticoagulant or/and antibody to phospholipid, such as anticardiolipin antibodies.

The unremarkable family history in this patient rules out an inherited thrombotic disorder. Normal TT rules out fibrinogen disorders. Prolonged PT and APTT in the absence of bleeding history eliminate the diagnosis of factor deficiency. The APTT test performed on a mixture of patient plasma and normal plasma did not correct the prolonged APTT. This result is indicative of an inhibitor. However, because the patient is not bleeding, factor VIII inhibitor is not indicated. A negative anticardiolipin antibody result rules out the possibility of anticardiolipin antibodies being responsible for the patient's clinical symptoms. The laboratory test result that confirms the presence of a lupus anticoagulant is prolonged APTT that is not corrected when mixed with normal plasma and that is neutralized by preincubation with platelet phospholipid (an excess of platelet phospholipid neutralizes the antibody, resulting in normal APTT).

15. **C** Performing liver biopsy in a patient with a prolonged PT and a high INR could have life-threatening consequences. In this patient, the prolonged PT is likely caused by liver disease. Vitamin K is stored in the liver and is essential for activation of factors II, VII, IX, and X. Vitamin K needs bile (secreted by the liver) for its absorption. In liver disease characterized by obstruction, bile is not secreted into the gastrointestinal tract, and therefore, vitamin K is poorly absorbed. The most logical course of action is the following: Start the patient on vitamin K therapy, repeat the PT test 4 days after starting vitamin K administration, and cancel the biopsy until the patient's PT returns to normal.

16. **A fresh blood sample was sent to the laboratory at 8:00 a.m. for the PT test. At 4:00 p.m., the doctor requested for the APTT test to be done on the same sample. What should the technologist do?**

- A. Rerun the APTT on the 8:00 a.m. sample and report the result
- B. Request a new sample for APTT
- C. Run the APTT in duplicate and report the average
- D. Mix patient plasma with normal plasma and run the APTT

Hemostasis/Select methods/Reagents/Specimen collection and handling/Specimens/3

17. **An APTT test is performed on a patient and the result is 50 sec (reference range 27–37 sec). The instrument flags the result because of failure of the delta check. The patient had had an APTT of 35 sec the previous day. The technologist calls the nursing unit to check whether the patient is on heparin therapy. The patient is not receiving heparin. What is the next appropriate step?**

- A. Check the patient's family history for inherited factor VIII deficiency
- B. Check to see if the patient has received any other anticoagulant medications
- C. Perform mixing studies
- D. Perform a factor VIII assay

Hemostasis/Select course of action/3

18. **A patient was put on heparin therapy postoperatively for prevention of thrombosis. The patient had the following laboratory results on admission: Platelet count = $350 \times 10^9/L$; PT = 12 sec (reference: 10–13 sec); APTT = 35 sec (reference: 28–37). After 6 days of heparin therapy, the patient complained of pain and swelling in her left leg. Her platelet count dropped to $85 \times 10^9/L$, and her APTT result was 36 sec. The physician suspected HIT and ordered the PLT aggregation test to be performed immediately. The heparin-**

induced PLT aggregation test result was negative. Heparin therapy was continued. Several days later, the patient developed a massive clot in her left leg that necessitated amputation. Which of the following should have been recognized or initiated?

- A.** The patient should have been placed on LMWH
- B.** The heparin dose should have been increased
- C.** The negative PLT aggregation does not rule out HIT
- D.** The patient should have been placed on warfarin therapy

Hemostasis/Correlate clinical and laboratory data/HIT/3

Answers to Questions 16–18

- 16. B** According to the CLSI guidelines, samples for APTT should be centrifuged and tested within 2 hours after collection. However, the sample is stable for 4 hours if stored at 4°C. APTT evaluates the clotting factors in the intrinsic and common coagulation pathways, including factor VIII (intrinsic) and factor V (common). Factors VIII and V are cofactors necessary for fibrin formation. However, they are both labile. Storage beyond 4 hours causes falsely elevated APTT results. The medical laboratory scientist should request a new sample for the APTT test.
- 17. B** Traditional anticoagulant drugs, such as heparin and warfarin, are well known. Newer anticoagulant drugs are available for the treatment and prevention of thrombosis. Some of these new drugs have AT or anti-factor Xa effects and, therefore, increase PT, APTT, and TT. Examples of these drugs are argatroban, which inhibits thrombin, and fondaparinux, which inhibits factor Xa.
- 18. C** Heparin therapy should be stopped immediately when clinical symptoms indicate HIT. The blood sample should be tested at least 4 hours after heparin therapy is discontinued. Early sampling for HIT testing may give a false-negative result because of the neutralization of antibody by heparin. LMWH should not be used in patients who develop HIT because LMWH can also cause HIT. Warfarin therapy can be started in patients who respond to heparin therapy. Heparin therapy must overlap warfarin therapy until the INR reaches a stable therapeutic range (2.0–3.0). Warfarin therapy could not be used in this patient because of lack of response to heparin therapy. The first step in the treatment of HIT is discontinuation of heparin, including intravenous catheter flushes, heparin-coated indwelling catheters, UFH, and LMWH.

19. A 50-year-old female was admitted to a hospital for hip replacement surgery.

Preoperative tests were performed, and the results showed the following:

Hgb = 13.5 g/dL; Hct = 42%; PT = 12 sec; APTT = 36 sec.

The patient was bleeding during surgery, and postoperative test results revealed the following Hgb = 5.0 g/dL; Hct = 16%; PT = 8 sec; APTT = 25 sec.

What steps should be taken before releasing these results?

- A.** No follow-up steps are needed; report the results as obtained
- B.** Report Hgb and Hct results, adjust the anticoagulant volume, and redraw a new sample for PT and APTT
- C.** Call the nurse and ask if the patient is receiving heparin
- D.** Because the patient is severely anemic, multiply the PT and APTT results by two and

report the results

Hemostasis/Select course of action/3

- 20. A 45-year-old woman visited her doctor complaining of easy bruising and menorrhagia occurring for the past few weeks. The patient had no history of excessive bleeding during child birth several years earlier or during a tonsillectomy in childhood. Her family history was unremarkable.**

Laboratory Tests	Patient	Reference Range
PT	45 sec	11–13 sec
APTT	125 sec	28–37 sec
TT	14 sec	10–15 sec
Mixing studies (patient plasma + normal plasma):	PT = 40 sec; APTT = 90 sec	
Platelet count and morphology:	normal	
Liver function tests:	normal	

These clinical manifestations and laboratory results are consistent with:

- A.** Factor VIII inhibitor
- B.** Factor V inhibitor
- C.** Factor VIII deficiency
- D.** Lupus anticoagulant

Hemostasis/Correlate clinical and laboratory data/Inhibitors/3

Answers to Questions 19–20

- 19. B** The anticoagulant:blood ratio should be adjusted for the PT and APTT tests in patients with a severe anemia. The standard anticoagulant volume (0.5 mL) is not sufficient for the large quantity of plasma in these patients, causing unreliable PT and APTT results. The low Hgb and Hct in this patient were caused by severe bleeding during surgery. To get accurate PT and APTT results, the amount of anticoagulant is adjusted according to the following formula: $(0.00185)(V)(100-H) = C$, where V = blood volume in mL; H = patient's Hct; and C = volume of anticoagulant in mL. A new sample should be drawn to rerun the PT and APTT. There are other causes for decreased PT and APTT, such as increased fibrinogen and increased factor VIII; however, the preanalytical variables affecting unreliable results should be ruled out first. Heparin therapy would increase PT and APTT.
- 20. B** The absence of a positive family history in this patient indicates acquired coagulopathy. Because both the PT and APTT test results are abnormal, the clotting factor involved is most probably in the common pathway. The lack of correction by mixing studies suggests the presence of an inhibitor. Factor V antibodies are the most common antibodies among the clotting factors of the common pathway (I, II, V, and X). Factor V antibodies are reported to be associated with surgery; some antibiotics,

such as streptomycin; exposure to blood products or the bovine form of “fibrin glue.” Patients with antibodies to factor V may require long-term therapy with immunosuppressive drugs. Acute bleeding episodes may be treated by platelet transfusions. The PT test is normal in patients with factor VIII deficiency and factor VIII inhibitor. Lupus anticoagulant is not present with bleeding unless associated with coexisting thrombocytopenia.

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